Nath 2021 HGSOC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) on 15 retained patients integrates the manifold: iLISI median 1.00 → 5.50 (of 15). Read on the regenerated embedding + per-neighbourhood iLISI — patients intermix across one connected component (a 3-cell speck removed). Mixed AND continuous.

Gate 2 — module geography → GO (non-branching)

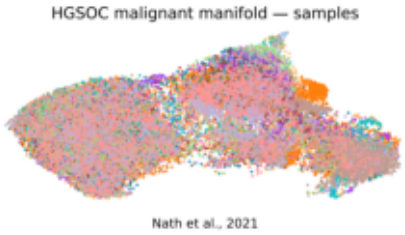
Over the low-cycling body a single (Gast/pSom) → Crest gradient: along pseudotime Gast/pSom fall (ρ −0.22 / −0.60) and Crest rises to a terminal end (ρ +0.21, Crest-high decile late). Crest opposes Gast in space (r=−0.41); joint pSom&Crest-high enrichment 0.42 (<1 = inverse gradient, not a co-expression hub).

Gate 3 — cross-repression → NA

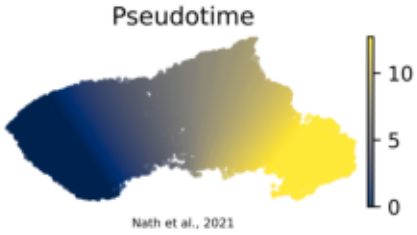
Not applicable — the manifold is non-branching (principal curve: 2 tips, 0 forks, fold-free). With no bifurcation there are no competing arms to test for mutual Gast↔pSom anti-correlation.

STEP FIGURES

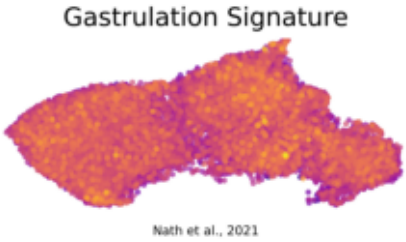
Manifold — samples



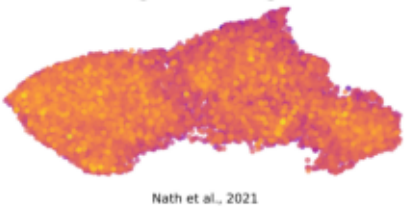
Pseudotime



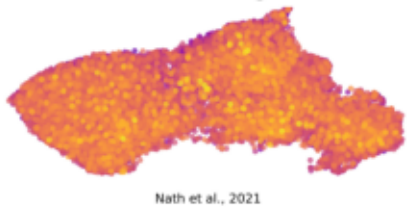
Gastrulation



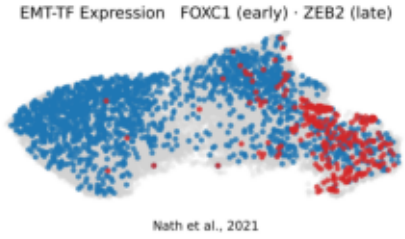
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
High-grade serous ovarian carcinoma (Nath et al., 2021), 10x — 32,484 malignant cells, 15 patients; trajectory scoped to the 26,703-cell low-cycling body. Non-branching Linear-pSom/Crest.

Verdict.
Pattern reproduces (non-branching). A single gastrulation/somitogenesis → neural-crest developmental axis over the low-cycling HGSOC body; no bifurcation. Geometry Linear-pSom/Crest. Gates G1 GO / G2 GO / G3 NA. EMT-TFs FOXC1 (early/root, 27%) → ZEB2 (terminus, 4%).

Parameter choices

Sample threshold	cliff 767→149 (ratio 5.15) → geom-mean 338; keeps 15/18 (drops iCell8 + P10)	**Integration**	Harmony on sample θ=10; n_pcs=20, n_neighbors=30, spread=1.3
Scope (cell-cycle fork)	proliferative lobe (5,781, 74% cyc) scoped out; body 26,703; modules cc-free	**Geometry / root**	non-branching curve (N=20, μ=100): 2 tips/0 forks; corr(t,x)=+0.98; root tip 0
Module means (raw)	Gast −0.004 · pSom +0.085 · Crest −0.010 (Linear-pSom/Crest)	**EMT-TFs**	FOXC1 (early/root, 27%) → ZEB2 (terminus, 4%); VIM excluded